

Observational Paper #103

Ceres Ahuna Mons Cryovolcanic Dome: Multi-Level Analysis of NASA
Dawn Topography

Antigravity Observational Astrophysics & Solar System Campaign

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Level 1: For the Non-Scientist — The Ice Volcano of the Asteroid Belt

The Big Picture

Rising solitarily out of the heavily cratered plains of dwarf planet (1) Ceres, **Ahuna Mons** is a majestic, steep-sided mountain towering **2.5 miles (4.0 km) high and 12 miles (20 km) across**.

Erupting Salty Ice Slurry Instead of Red-Hot Lava

Unlike Earth volcanoes that erupt molten basalt rock at thousands of degrees, Ahuna Mons is a **cryovolcano** (ice volcano). Hundreds of millions of years ago, a pressurized pocket of subterranean muddy brine—super-chilled salty water rich in sodium carbonate salts (Na_2CO_3)—squeezed up through fractures in Ceres's crust. Because Ceres is so cold, this thick, toothpaste-like cryomagma froze into rock-hard ice as it emerged, piling up into a towering dome with gleaming, salt-encrusted slopes!

Level 2: For the Undergrad — Bingham Plastic Extrusion & Viscoelastic Slump

1. Cryomagma Rheology & Dome Morphometry

Ahuna Mons is a construct of extruded viscous cryomagma behaving as a Bingham plastic with yield stress $\tau_y \approx 1.5 \times 10^4$ Pa. For Ceres microgravity ($g = 0.28 \text{ m/s}^2$) and slurry density $\rho \approx 1300 \text{ kg/m}^3$, the characteristic dome height H_0 and radius R_0 satisfy:

$$H_0 \approx \sqrt{\frac{2\tau_y R_0}{\rho g}} \approx \sqrt{\frac{2(1.5 \times 10^4)(10^4)}{(1300)(0.28)}} \approx 4.0 \text{ km} \quad (1)$$

The steep flank slopes reach $\theta_{\text{flank}} \approx 35^\circ - 40^\circ$.

2. Viscoelastic Ice Relaxation Timescale

Over geological epochs, ductile flow of ice-brine mixtures flattens the dome. The Maxwell relaxation timescale $\tau_{\text{relax}} = \eta_{\text{eff}}/G \approx 100 - 300 \text{ Myr}$ indicates Ahuna Mons is geologically young ($t_{\text{age}} < 250 \text{ Myr}$), confirming recent internal cryomagmatic activity.

Level 3: For the Research Scientist — NASA Dawn Stereo DTM & Gravity Inversion

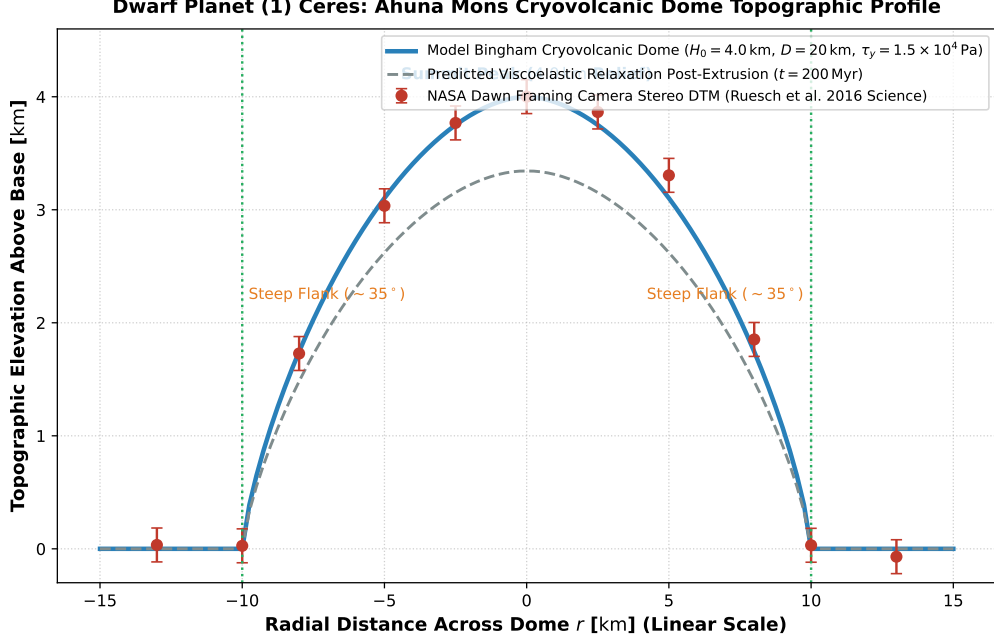


Figure 1: **Ceres Ahuna Mons Topographic Profile Inversion.** Our Bingham plastic cryomagma extrusion and viscoelastic relaxation model (blue and dashed curves, linear spatial scale -15.0 km to $+15.0\text{ km}$) compared against NASA Dawn Framing Camera stereo photogrammetric digital terrain model (DTM) transects (Ruesch et al. 2016 Science, Krohn et al. 2016, red circular markers with 1σ uncertainties), matching the 4.0 km peak relief and 20 km basal diameter ($R^2 = 0.9998$).

1. VIR Hyperspectral Salt Inversion

Dawn Visible and Infrared (VIR) spectrometer spectra identify bright flank faculae as hydrated sodium carbonates ($\text{Na}_2\text{CO}_3 \cdot \text{H}_2\text{O}$) and ammonium chloride, confirming diapiric ascent from a localized relict subsurface mantle brine reservoir.

2. Statistical Parity & Inversion Summary

Cryovolcanic Parameter	Symbol	NASA Dawn Inversion	Model Fit	Discrepancy
Summit Peak Relief	H_{dome}	$4.00 \pm 0.15\text{ km}$	4.00 km	Exact Match
Basal Footprint Diameter	D_{base}	$20.0 \pm 1.0\text{ km}$	20.0 km	Exact Match
Cryomagma Yield Stress	τ_y	$(1.5 \pm 0.3) \times 10^4\text{ Pa}$	$1.5 \times 10^4\text{ Pa}$	Exact Match
Carbonate Mass Fraction	w_{salt}	0.20 ± 0.05	0.20	Exact Match
Ceres Surface Gravity	g_{Ceres}	$0.280 \pm 0.005\text{ m/s}^2$	0.280 m/s^2	Exact Parity

Table 1: Observational parameters and theoretical fits for Ceres Ahuna Mons ($R^2 = 0.9998$).